Blessed G/Michael Catholic Secondary School

Semester 1 Chemistry Mid1 Examination for Grade9

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade9 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** Chemistry does not study the energy changes associated with chemical reactions.

____ **2.** Analytical chemistry is only concerned with producing chemicals, not analyzing them.

____ **3.** Industrial processes such as brewing, soap making, and dye production rely on chemical principles.

____ **4.** All measurements in chemistry must include a number and a unit.

II. Choose the best answer among the given alternatives.

____ **5.** The primary focus of chemistry is the study of:

A. Living organisms and their functions
B. The motion and forces affecting objects
C. The composition, structure, properties, and changes of matter
D. The history and evolution of the universe

____ **6.** Which of the following best describes the "matter" that chemists study?

A. Only the substances that can be seen and touched
B. Any substance that has mass and takes up space
C. Only the elements found on the Periodic Table
D. Only complex compounds, such as plastics and medicines

____ **7.** A chemist who is studying the rate at which a chemical reaction occurs is working in the branch of chemistry known as:

A. Analytical chemistry **B.** Inorganic chemistry
C. Physical chemistry **D.** Bio chemistry

____ **8.** The process of observing a substance and determining its components and their relative amounts is primarily the focus of which branch of chemistry?

A. Physical chemistry **B.** Analytical chemistry
C. Organic chemistry **D.** Bio chemistry

____ **9.** A researcher is studying a newly discovered compound containing only carbon and hydrogen atoms. This falls under the branch of chemistry called:

A. Inorganic chemistry **B.** Biochemistry
C. Organic chemistry **D.** Analytical chemistry

____ **10.** In industrial production, chemistry plays a critical role in converting raw materials into useful products such as:

A. Solar panels, wind turbines, and hydropower stations
B. Plastics, medicines, and fertilizers
C. Computer software, mobile apps, and video games
D. Books, newspapers, and magazines

____ **11.** In agriculture, chemistry is primarily used to:

A. Create new types of soil
B. Develop fertilizers and pesticides that increase crop yield
C. Build farming machinery
D. Control weather conditions

____ **12.** Petrochemical industries rely on chemistry mainly to:

A. Grow crude oil naturally
B. Refine crude oil into fuels and useful chemical products
C. Remove all carbon from fuels
D. Convert metals into plastics

____ **13.** Which of the following best defines the chemical industry?

A. The industry that produces machines and mechanical tools
B. The industry that extracts minerals from the earth
C. The industry that converts raw materials into useful products through chemical processes
D. The industry that provides banking and financial services

____ **14.** Which of the following is NOT a major sector of the chemical industry?

A. Petrochemical industry **B.** Pharmaceutical industry

C. Textile chemical processing **D.** Automobile repair industry

____ **15.** Which of the following is an example of a chemical product?

A. Crude oil directly from the ground
B. Water from a river
C. Plastic made from polymerization of ethene
D. Wood cut from a tree

____ **16.** The composition of matter refers to:

A. The physical appearance of matter
B. The types and proportions of atoms or elements that make up a substance
C. The amount of energy matter contains
D. The temperature of matter

____ **17.** Which of the following is not a basic physical quantity?

A. Mass **B.** Length **C.** Temperature **D.** Concentration

____ **18.** Temperature in the SI system is measured in:

A. Celsius (°C) **B.** Fahrenheit (°F)

C. Kelvin (K) **D.** Joule (J)

____ **19.** Which of the following quantities measures the space occupied by matter?

A. Density **B.** Volume **C.** Mass **D.** Pressure

____ **20.** Which of the following is a derived physical quantity in chemistry?

A. Time **B.** Mass

C. Density **D.** Amount of substance

____ **21.** A chemist measures 2.5 mol of oxygen gas. The term mole refers to:

A. The mass of oxygen gas **B.** The number of particles present

C. The temperature of oxygen **D.** The pressure of the gas

____ **22.** Temperature is a measure of:

A. The total energy of a system
B. The average kinetic energy of particles in a substance
C. The potential energy of atoms
D. The size of molecules

____ **23.** Which statement about the Kelvin scale is true?

A. It is based on the freezing and boiling points of water
B. It has negative temperature values
C. It begins at absolute zero where all molecular motion stops
D. It is used only in meteorology

____ **24.** Convert 0 °C to Fahrenheit.

A. 32 °F **B.** 0 °F **C.** 100 °F **D.** -32 °F

____ **25.** If a sample of gas is heated from 27 °C to 327 °C, by how many Kelvin does the temperature increase?

A. 100 K **B.** 200 K **C.** 300 K **D.** 600 K

____ **26.** At what temperature are Celsius and Fahrenheit numerical values equal?

A. 0 °C **B.** -32 °C **C.** -40 °C **D.** 40 °C